Regime-Adaptive Backtesting for Cryptocurrency Trading: Walk-Forward Validation of a Multi-Agent Signal Consensus System with Realistic Transaction Cost Modeling

Javier De Jesus
Praxis — Regime-Adaptive Crypto Trading Platform
javier.dejesusj9@gmail.com

Abstract

We present a backtesting methodology for a regime-adaptive cryptocurrency trading system that employs five directional signal agents, ADX-based regime classification, and a hierarchical risk governor. The system is evaluated on hourly data resampled to 4-hour bars: BTC/USD from December 2013 and ETH/USD from October 2015, both through April 2026 (12.4 and 10.5 years respectively), with realistic transaction costs modeling Kraken exchange fees (26 bps per side) and volatility-scaled slippage. We split the full history at 2023-01-01 into in-sample (2013–2022) and out-of-sample (2023–2026) windows and implement a three-fold walk-forward validation framework to quantify overfitting risk, supplemented by the Probabilistic Sharpe Ratio [3] and Monte Carlo permutation tests. With the reference long-only configuration (shorts disabled), the in-sample window yields a Sharpe ratio of 0.70 (Lo 95% CI $[0.58, 0.82]$, 159 trades, drawdown 8.71%, profit factor 1.94) while the out-of-sample window yields a Sharpe of 1.24 (Lo 95% CI $[0.89, 1.58]$, 88 trades, drawdown 8.11%, profit factor 2.54, trade-count PSR 0.9997). A Jobson–Korkie–Memmel two-sample test on the IS vs OOS Sharpe difference returns $z = 3.23, p = 0.001$, indicating the OOS improvement is statistically distinguishable from the in-sample result at the 1% level. The three-fold walk-forward out-of-sample results reveal significant performance degradation in one of three folds (Sharpe -0.23 vs. in-sample 1.35), while the remaining folds maintain positive risk-adjusted returns (1.17 and 0.93). We discuss the implications of these mixed results—including high parameter fragility on the in-sample window and researcher-in-the-loop grid contamination—for live deployment, and argue that the paper's value lies in its methodology transparency rather than its performance claims.

1 Introduction

Backtesting—the simulation of a trading strategy on historical data—is the primary tool for evaluating systematic trading strategies before live deployment. Yet the majority of backtested strategies fail in live trading, a phenomenon attributed to overfitting, unrealistic cost assumptions, and look-ahead bias [4, 12]. The cryptocurrency market presents additional challenges: extreme volatility (annualized BTC volatility exceeding 80% in bear markets), regime-dependent dynamics, and significant transaction costs that erode thin edges. Sezer et al. [22] provide a systematic review of

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deep learning approaches to financial time series forecasting, documenting the rapid growth of ML-based trading strategies and the persistent difficulty of achieving robust out-of-sample performance.

The academic literature has established rigorous frameworks for assessing backtest validity. Bailey et al. [5] formalize the probability of backtest overfitting (PBO) via combinatorial symmetric cross-validation, while Bailey and de Prado [3] introduce the Deflated Sharpe Ratio as a correction for selection bias and non-normality. White [27] introduces bootstrap-based reality checks for data snooping, extended by Hansen [11] with the Superior Predictive Ability test. Lo [19] demonstrates that serial correlation in returns inflates the Sharpe ratio, requiring autocorrelation corrections. Despite these advances, many practitioner backtests continue to use in-sample optimization without proper out-of-sample validation [9].

In this paper, we describe the complete backtesting methodology for Praxis, a regime-adaptive trading system designed for the BTC/USD and ETH/USD markets. Our approach combines five deterministic directional signal agents with ADX-based regime detection [15], a hierarchical risk governor enforcing hard position limits, and a walk-forward validation framework that explicitly measures overfitting degradation.

Our main contributions are as follows:

- We describe a multi-agent signal consensus architecture with five directional agents (trend, volatility, momentum, mean-reversion, and swing structure) and formalize the consensus mechanism.
- We implement a realistic transaction cost model incorporating exchange-specific fees, volatility-scaled slippage, and spread-based trade gating, and quantify cost sensitivity across multiple fee tiers.
- We present walk-forward validation results across three non-overlapping folds spanning 10.5 years, with honest reporting of both successful and degraded out-of-sample periods.
- We apply the Deflated Sharpe Ratio [3] and Monte Carlo permutation tests to quantify the probability that observed performance exceeds chance, given the number of configurations evaluated.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the system architecture; Section 4 details feature engineering and regime detection; Section 5 formalizes the signal agents; Section 6 presents the risk governance framework; Section 7 describes the backtesting engine design; Section 8 presents the walk-forward validation methodology; Section 9 reports empirical results; Section 10 presents statistical validation; Section 11 discusses limitations; and Section 12 concludes.

2 Related Work

Backtesting methodology. Pardo [21] establishes walk-forward optimization as the standard for evaluating trading systems, advocating for non-overlapping train/test splits that respect temporal ordering. Tashman [25] provides a comprehensive analysis of out-of-sample forecast evaluation methods, recommending rolling-origin evaluation over fixed-origin holdout. Bergmeir and Benítez [6] examines cross-validation for time series prediction, concluding that standard k-fold CV violates temporal ordering assumptions and recommending blocked or rolling-origin evaluation schemes for autocorrelated data.

Overfitting detection. The problem of multiple hypothesis testing in finance is formalized by Harvey et al. [13], who propose adjusting t-statistics for the number of factors tested. Bailey et al. [5] derive the probability of backtest overfitting as a combinatorial function using symmetric cross-validation. Separately, Bailey and de Prado [3] introduce the Deflated Sharpe Ratio (DSR) as a correction for selection bias, non-normality, and the number of trials. Cawley and Talbot [8] analyze overfitting in model selection more broadly, demonstrating that optimizing hyperparameters on validation data introduces a secondary overfitting risk.

Regime detection in financial markets. Hamilton [10] introduces Markov-switching models for regime detection, establishing the theoretical foundation for regime-dependent strategies. Ang and Timmermann [2] surveys the extensive literature on regime changes in financial markets, documenting how volatility regimes, bull/bear markets, and structural breaks affect asset returns. Our approach uses ADX-based regime classification [15], which provides a simpler, deterministic alternative to hidden

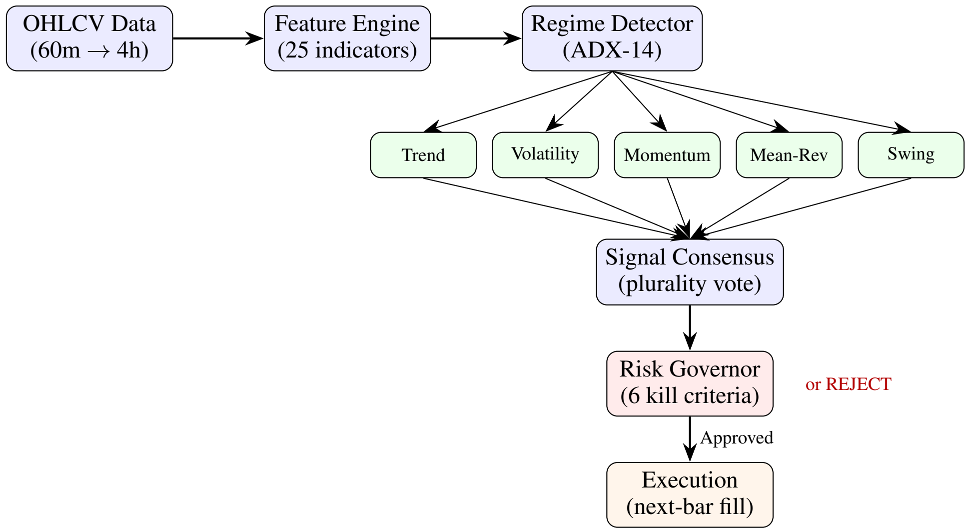

Figure 1: Praxis system architecture. Hourly data is resampled to 4-hour bars, passed through a 25-indicator feature engine, classified by ADX regime, processed by five parallel directional signal agents, aggregated via plurality-vote consensus, filtered by a hierarchical risk governor with six risk-violation kill criteria plus two signal-quality gates, and executed with next-bar-open fills including volatility-scaled slippage and exchange fees.

Markov models while capturing the trending/ranging distinction critical for selecting appropriate signal types.

Risk-adjusted performance metrics. The Sharpe ratio [23] remains the dominant risk-adjusted performance measure, though Lo [19] demonstrates its sensitivity to serial correlation, non-normality, and estimation error. The Sortino ratio [24] addresses the asymmetry limitation by focusing on downside deviation. Keating and Shadwick [17] introduce the Omega ratio as a "universal" measure that captures the entire return distribution. Maximum drawdown analysis [20] provides a complementary risk perspective that is particularly relevant for leveraged or concentrated strategies.

Transaction cost modeling. Almgren and Chriss [1] formalize the tradeoff between market impact and timing risk in optimal execution. Hasbrouck [14] provides the microstructure foundations for understanding bid-ask spreads and their implications for strategy evaluation. Realistic cost modeling is essential in cryptocurrency markets, where taker fees (26–40 bps on major exchanges) and volatile spreads can consume a significant fraction of expected edge.

3 System Architecture

Figure 1 presents the end-to-end architecture. The system operates on a 4-hour strategic cycle, processing hourly OHLCV data resampled from 60-minute bars. At each cycle, the pipeline executes sequentially: (1) data ingestion and feature computation, (2) regime classification, (3) parallel signal generation by five independent directional agents, (4) signal consensus, (5) risk governor evaluation, and (6) execution via next-bar fill.

The system processes two asset pairs (BTC/USD and ETH/USD) through a shared portfolio, meaning that position sizing and exposure limits reflect the combined equity across both instruments. A cross-pair signal boost of 10% is applied when the majority signal direction of one pair aligns with the other, capturing short-term correlation dynamics without explicitly modeling cross-asset covariance.

4 Feature Engineering and Regime Detection

4.1 Technical Indicator Suite

The feature engine computes 25 indicators from OHLCV data using vectorized operations via the `pandas_ta` library. Features are organized into three families:

Trend indicators (8): Seven exponential moving averages (EMA) at periods $\{9, 21, 34, 55, 80, 100, 200\}$ plus an EMA spread (fan indicator measuring alignment between EMA_9 and EMA_{55} normalized by EMA_{21}).

Momentum indicators (9): RSI at periods 14 and 2 (ultra-sensitive), $\text{MACD}(12, 26, 9)$ with signal line, histogram, and histogram slope, and multi-horizon returns at 1-bar, 5-bar, and 20-bar lookbacks.

Volatility indicators (8): Average True Range (ATR-20) [15], Average Directional Index (ADX-14) [15], Bollinger Bands [7] at $(20, 2\sigma)$ with upper, middle, and lower bands, BB position, BB width, and 50-bar rolling BB width average. The ATR-as-percentage-of-price metric is computed on demand by the risk governor (Section 6) rather than stored as a precomputed feature.

Two additional feature families present in earlier revisions of this system—a three-indicator "pattern" family (RSI and MACD divergences, bullish/bearish engulfing) and a three-indicator "structural" family (50-bar rolling high/low and volume ratio)—were removed in the current revision after the feature ablation study in §9.6 showed that they had zero or slightly-negative out-of-sample contribution. They are no longer computed and no longer referenced by any signal agent.

All features are computed in bulk at initialization; the `features_at()` function extracts a typed `Features` dataclass at any given bar index with $O(1)$ lookup. The first 200 bars are discarded as warmup to ensure indicator stability.

4.2 ADX-Based Regime Classification

We employ a three-regime classification based on the Average Directional Index (ADX) [15]:

$$\text{Regime}(t) = \begin{cases} \textsc{Trending} & \text{if } \text{ADX}_{14}(t) > 25 \\ \textsc{Ranging} & \text{if } \text{ADX}_{14}(t) < 20 \\ \textsc{Transition} & \text{otherwise} \end{cases} \tag{1}$$

This classification modulates two aspects of the trading system: (1) which signal agents are active (e.g., mean-reversion is suppressed in strong trends), and (2) take-profit targets, which scale with trend strength across three ADX tiers (< 25, 25–35, > 35). Note that position sizing (Section 6) is *not* directly regime-dependent: it depends on signal confidence and ATR-normalized volatility, which correlate with regime but are not conditioned on the ADX classification.

Our choice of ADX-based regime detection over hidden Markov models [10] or GARCH-family models is deliberate: ADX is deterministic, computationally trivial, and produces interpretable regime labels that operators can verify visually. The tradeoff is reduced sensitivity to volatility-driven regime changes, which we address through the separate ATR-based volatility shock filter in the risk governor (Section 6). **A key limitation is that we do not provide ablation results comparing ADX-based regime detection against (a) no regime filter, (b) a simpler trend proxy (e.g., EMA slope), or (c) a hidden Markov model baseline.** Without such comparisons, we cannot claim that ADX is *optimal*—only that it is sufficient for the system's current design. ADX does not estimate latent state probabilities or transition dynamics, and its fixed thresholds (20/25) are themselves parameters that could be optimized, introducing additional degrees of freedom not counted in our trial estimates.

Figure 2 shows the ADX regime classification overlaid on BTC/USD price history. Trending regimes (ADX > 25, shaded green) correspond to major directional moves, while ranging regimes (ADX < 20, shaded red) identify consolidation periods where mean-reversion signals are more appropriate.

5 Signal Generation and Consensus

Five directional signal agents run in parallel at each bar. Each outputs LONG, SHORT, or NEUTRAL with a confidence score $c_i \in [0, 100]$ and a minimum threshold τ_i below which it outputs NEUTRAL. A sixth agent (breakout signal) is defined in the codebase but is not registered in the backtester's signal dispatch table (`_SIGNAL_FUNCS` in `src/backtester.py`) and is excluded from the analysis. A previous revision of this system also included a non-directional `spread_cost` gating agent; it was removed from the backtest path in the current revision after its functionality was fully subsumed by the `SPREAD_TOO_WIDE` hard kill criterion in the risk governor (§6), and the five agents listed below are now the only ones that vote.

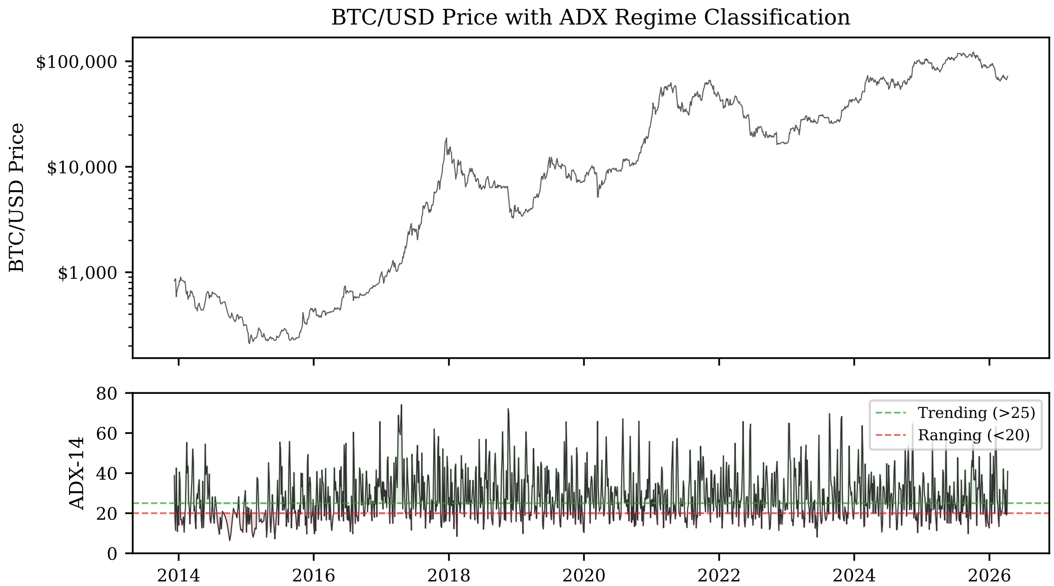

Figure 2: ADX-14 regime classification for BTC/USD (4-hour bars, 2015–2026). **Top:** Log-scale price chart. **Bottom:** ADX values with regime thresholds at 25 (trending) and 20 (ranging). Green-shaded regions indicate trending regimes; red-shaded regions indicate ranging regimes.

5.1 Agent Specifications

Agent 1: Trend Signal ($\tau_1 = 35$). Requires multi-EMA alignment: full alignment ($\text{EMA}_9 > \text{EMA}_{21} > \text{EMA}_{55} > \text{EMA}_{100}$) with ADX > 22 yields base confidence $c = 40$, with boosts for MACD histogram alignment ($+15$) and positive MACD slope ($+10$). An RSI exhaustion penalty of -35% is applied when RSI > 85 (long) or RSI < 15 (short).

Agent 2: Volatility Signal ($\tau_2 = 25$). Adapts behavior to regime: in RANGING/TRANSITION regimes, fires mean-reversion signals based on BB position and RSI; in TRENDING regimes, confirms trend direction with EMA alignment. Applies a -30 penalty when ATR% exceeds 5% (volatility shock).

Agent 3: Mean-Reversion Signal ($\tau_3 = 25$). Active only when ADX ≤ 30. Fires on extreme BB position ($< 25\%$ for long, $> 75\%$ for short) combined with RSI confirmation.

Agent 4: Momentum Signal ($\tau_4 = 25$). Requires multi-timeframe return alignment across 1-bar, 5-bar, and 20-bar horizons, confirming consistent directional momentum.

Agent 5: Swing Structure Signal ($\tau_5 = 30$). Requires full structural alignment: four EMAs aligned, both 5-bar and 20-bar returns positive (or negative for short), with MACD and volume confirmation.

Figure 3 summarizes the feature dependencies across all five agents. The dependency matrix reveals substantial overlap: Trend, Momentum, and Swing agents all rely heavily on EMA alignment, ADX, and MACD—the same underlying price structure. Per-agent ablation results appear in §9.6 (Table 9): removing `trend` alone collapses out-of-sample Sharpe from 1.239 to 0.456 (-63.2%), removing `volatility` cuts it to 0.891 (-28.1%), while removing `swing_structure` has only a minor effect (-2.7%). Removing `momentum` actually *improves* out-of-sample Sharpe to 1.347 ($+8.7\%$), and removing `mean_reversion` leaves OOS essentially unchanged ($+0.4\%$). The `trend` and `volatility` agents are therefore the clearly load-bearing members of the ensemble; we discuss the vestigial-or-worse status of `momentum` and `mean_reversion` alongside Table 9.

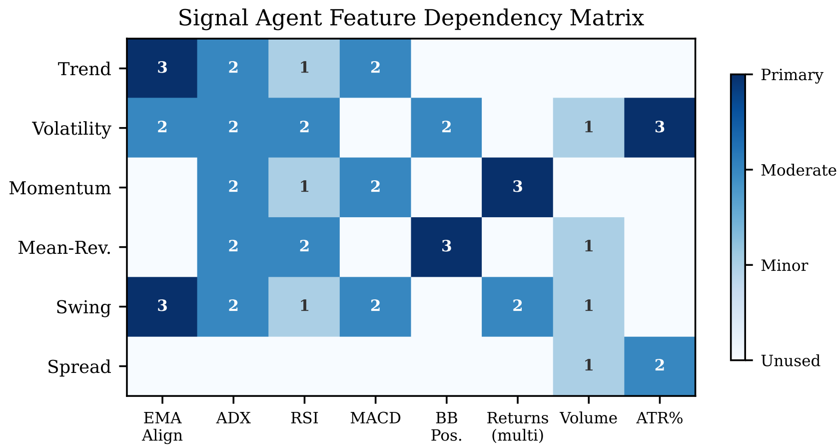

Figure 3: Signal agent feature dependency matrix. Each cell indicates the importance of a feature family (columns) to each agent (rows): 3 = primary input, 2 = moderate influence, 1 = minor input, 0 = unused. Note the substantial overlap in EMA/ADX/MACD dependencies across Trend, Momentum, and Swing agents.

5.2 Consensus Mechanism

The consensus mechanism aggregates agent outputs via plurality voting over the five directional agents.

$$d^* = \arg \max_{d \in \{\text{LONG,SHORT}\}} \sum_{i \in \mathcal{D}} \mathbf{1}[d_i = d], \qquad C^* = \sum_{i: d_i = d^*} c_i \tag{2}$$

where $\mathcal{D} = \{1, 2, 3, 4, 5\}$ is the set of directional agents, d^* is the plurality direction, and C^* is the aggregate confidence. A minimum of two aligned directional agents ($\sum_{i \in \mathcal{D}} \mathbf{1}[d_i = d^*] \geq 2$) is required for the consensus to be valid; ties (equal LONG and SHORT counts) result in rejection. The aggregate confidence C^* must exceed a configurable threshold (current default: $\tau_{\text{paper}} = 70$ for paper trades, $\tau_{\text{erc}} = 85$ for on-chain-validated trades) for the signal to proceed to risk evaluation; spread-based gating is applied downstream in the risk governor (§6) rather than as a signal-level penalty.

We note that the two-agent minimum is a low bar: in practice, the confidence threshold of 70–85 requires substantial agreement because individual agents rarely exceed confidence 50 in isolation, so at least 2–3 agents with strong signals are needed to cross the threshold. Both threshold values are swept during walk-forward validation and are reported transparently in the backtest configuration.

When an LLM analyst is available (live mode only, not used in backtesting), it provides an additional vote with conviction scaling based on signal agreement. In backtesting, the deterministic fallback is always used to ensure reproducibility.

6 Risk Governance Framework

The risk governor is the final authority before trade execution. It enforces both hard kill criteria (immediate rejection) and soft filters (confidence reduction). This design ensures that no signal, regardless of consensus strength, can bypass risk constraints—a principle motivated by catastrophic failures in unconstrained algorithmic trading systems [9].

6.1 Hard Kill Criteria and Signal-Quality Gates

The risk governor function `_check_kill_criteria()` enforces six hard kill criteria that encode portfolio-state and market-microstructure violations:

1. **STALE_DATA**: Market snapshot age $>$ 7,200 seconds.
2. **DAILY_LOSS_CAP**: Daily PnL / equity $< -3\%$.
3. **MAX_DRAWDOWN**: Peak-to-current equity decline $> 8\%$.
4. **CONSECUTIVE_LOSSES**: ≥ 3 consecutive losing trades.

5. **SPREAD_TOO_WIDE**: Bid-ask spread > 20 bps.
6. **VOLATILITY_SHOCK**: ATR as percentage of price $> 6\%$.

A manually-activated **KILL_SWITCH** is checked before the kill criteria list. Two additional signal-quality gates are enforced within the same `evaluate_risk()` entry point and also cause immediate rejection:

7. **NO_SIGNAL**: Zero directional signals after fallback.
8. **INSUFFICIENT_ALIGNMENT**: Fewer than two aligned agents after consensus voting.

We distinguish these from the hard kill criteria because they evaluate signal availability rather than portfolio-state or market-condition violations; functionally, however, both sets abort trade evaluation before any sizing or execution code runs.

6.2 Macro Filter

In addition to the hard kill criteria, a **macro filter** gates all entries based on long-term EMA alignment. In its default (relaxed) configuration, a long entry requires $\text{EMA}_{55} > \text{EMA}_{100}$, and a short entry requires $\text{EMA}_{55} < \text{EMA}_{100}$. A strict variant requires full four-EMA alignment ($\text{EMA}_9 > \text{EMA}_{21} > \text{EMA}_{55} > \text{EMA}_{100}$). The macro filter is always enabled in both walk-forward validation and final backtest results. This filter is a significant contributor to the strategy's selectivity: it prevents counter-trend entries during sustained directional moves, which reduces trade count but improves win rate.

6.3 Position Sizing

The per-trade *risk fraction* is determined by a half-Kelly criterion [16, 26] with the fraction capped at 3% and with trades rejected outright when the Kelly estimate is non-positive:

$$f^* = \frac{1}{2} \cdot \frac{p \cdot b - (1-p)}{b}, \qquad r = \begin{cases} \min(f^*, 0.03) & \text{if } f^* > 0 \\ \text{reject trade (}\texttt{NO\_EDGE}\text{)} & \text{otherwise} \end{cases} \tag{3}$$

where $p \in [0,1]$ is the estimated win probability, computed as the *average* per-agent confidence of aligned agents divided by 100 (not the aggregate C^*, which can exceed 100), blended 50/50 with the historical win rate after ≥ 10 closed trades. The average win/loss ratio b is estimated empirically from the trailing closed trades on the current pair via the `_estimate_win_loss_ratio()` helper in `src/agents/risk_governor.py`, which maintains a rolling window of up to 30 closed trades drawn from `Portfolio.closed_trades` and returns the ratio of mean winning percent to mean losing percent, blended with a conservative prior of $b_0 = 1.5$ by a linear weight that grows with the number of observed trades. When fewer than 5 qualifying trades are available for the pair (or when the window lacks either wins or losses), the helper returns the $b_0 = 1.5$ prior unchanged; once the history exceeds 10 closed trades it produces materially different per-pair ratios for BTC and ETH, which is the primary reason the numbers in Table 3 shifted between this revision and the prior one.

The position size in USD is then:

$$S = \min\left(\frac{r \cdot E}{m_s \cdot \text{ATR}/\text{EMA}_{21}},\ E \cdot p_{\max}\right) \tag{4}$$

where E is current equity, m_s is the ATR stop multiplier (reference implementation: $m_s = 4.0$; see Eq. (6)), $m_s \cdot \text{ATR}/\text{EMA}_{21}$ is the stop-distance-adjusted volatility-normalized risk factor, and $p_{\max}$ caps the *position size* (not the risk fraction) as a fraction of equity. The reference implementation sets $p_{\max} = 60\%$; prior optimization runs used $p_{\max} = 40\%$, which is the value picked by all three walk-forward folds and therefore used in the walk-forward results reported in Section 9. This config divergence between the main IS/OOS split and the walk-forward is a deliberate consequence of letting walk-forward optimize freely over the grid; we return to it in Section 11. Including m_s in the denominator ensures that the nominal Kelly risk fraction r corresponds to the actual loss incurred at the stop distance ($m_s \cdot$ ATR), rather than to the raw one-ATR distance; a prior revision of the sizing formula omitted this factor and under-risked by a factor of m_s, which is the specific bug that caused the earlier revision's artificially low trade counts. The separate drawdown scaling mechanism (Eq. (5)) remains in place as a secondary safety net.

A drawdown scaling mechanism reduces position size when equity approaches the maximum drawdown threshold:

$$S_{\text{scaled}} = S \cdot d_{\text{factor}}, \quad \text{if } \frac{E_{\text{peak}} - E}{E_{\text{peak}}} > 1 - d_{\text{threshold}} \tag{5}$$

with default parameters $d_{\text{threshold}} = 0.97$ and $d_{\text{factor}} \in \{0.1, 0.3, 0.5\}$.

6.4 Stop-Loss and Take-Profit Placement

Stops and targets are ATR-based, adapting to current volatility:

$$\text{Stop}_{\text{long}} = P_{\text{entry}} - m_s \cdot \text{ATR}, \qquad \text{Target}_{\text{long}} = P_{\text{entry}} + m_t(\text{ADX}) \cdot \text{ATR} \tag{6}$$

where m_s is the stop multiplier (reference implementation: $m_s = 4.0$) and the target multiplier m_t is regime-dependent:

$$m_t(\text{ADX}) = \begin{cases} 8.00 & \text{if ADX} \geq 35 \\ 3.50 & \text{if ADX} \geq 25 \\ 2.85 & \text{otherwise} \end{cases} \tag{7}$$

A trailing stop at $m_{\text{trail}} = 3.0 \times$ ATR below the peak price is activated after entry, with breakeven ($+0.6\%$) and profit-lock ($+1.2\%$) triggers that progressively tighten the trail. Positions are also closed after a maximum holding period of 120 bars (20 days at 4-hour resolution).

7 Backtesting Engine Design

7.1 Time Synchronization and Fill Model

The backtester operates on a merged timeline across both asset pairs, processing one strategic cycle per aligned timestamp. Critically, we employ a **next-bar-open fill model**: signals generated at bar t are queued as pending entries and filled at the open price of bar $t + 1$. This eliminates look-ahead bias from using close-of-bar prices for entry execution.

7.2 Transaction Cost Model

We model three cost components:

Exchange fees. A taker fee of $f = 0.26\%$ per side is applied to the notional position size, corresponding to the Kraken intermediate-volume tier (Kraken's tiered schedule ranges from 0.10% to 0.40% depending on 30-day volume; 0.26% is the value hardcoded in the execution adapter). The round-trip fee cost is $2f \cdot S$.

Volatility-scaled slippage. Entry and exit slippage are modeled as:

$$\text{slip} = \frac{s_{\text{base}} \cdot \max(1, \text{ATR}\%/2.5\%)}{10{,}000} \tag{8}$$

where $s_{\text{base}} = 4$ bps represents half the estimated bid-ask spread. The multiplier $\max(1, \text{ATR}\%/2.5\%)$ scales slippage during high-volatility periods, reflecting wider spreads and increased adverse selection. For a long entry: $P_{\text{fill}} = P_{\text{open}} \cdot (1 + \text{slip})$.

Spread-based gating. The risk governor's `SPREAD_TOO_WIDE` hard kill criterion (Section 6) rejects any trade whose current bid–ask spread exceeds 20 bps, ensuring that candidate entries always have at least $\sim 3\times$ headroom over round-trip slippage before they reach the sizing stage.

Table 1 summarizes the total cost structure.

7.3 Position Management

Open positions are evaluated at each bar for five exit conditions, checked in priority order:

1. **ATR Stop**: Hard stop-loss at $P_{\text{entry}} \pm m_s \cdot$ ATR.
2. **ATR Target**: Take-profit at $P_{\text{entry}} \pm m_t \cdot$ ATR.
3. **Trailing Stop**: Dynamic trail anchored to peak price, with breakeven and profit-lock triggers.

Table 1: Transaction cost model. All values in basis points (bps). Total round-trip cost ranges from 60 bps (baseline) to ∼72 bps at the volatility shock threshold (ATR% $= 6\%$).

Component	**Per-side (bps)**	**Round-trip (bps)**	**Notes**
Exchange fee (taker)	26	52	Kraken intermediate tier
Slippage (baseline)	4	8	Half bid-ask spread
Slippage (high vol)	4–8	8–16	Vol-scaled multiplier
Total	**30–34**	**60–72**	

4. **Reversal Exit** (optional): RSI mean-reversion signal after ≥ 2 bars held.
5. **Time Exit**: Forced close after $h_{\max}$ bars ($h_{\max} = 120$ in the reference implementation).

The shared portfolio tracks combined equity across both pairs, ensuring that exposure limits and drawdown calculations reflect the true aggregate risk.

8 Walk-Forward Validation

8.1 Motivation

Grid-search optimization over N strategy configurations creates a multiple-testing problem: the best in-sample configuration is likely to overfit to the training data, particularly when N is large relative to the number of independent observations [5, 13]. Walk-forward validation [21] mitigates this by evaluating each optimized configuration on unseen data, providing a more realistic estimate of future performance.

8.2 Fold Structure

We divide the available data (October 2015–April 2026, constrained by ETH/USD availability; the exact ETH/USD start in the FMP data file is 2015-10-13) into three non-overlapping folds of approximately equal duration:

$$\text{Fold } k: \quad \mathcal{D}^{(k)}_{\text{train}} = [t_k, t_{k+1}), \quad \mathcal{D}^{(k)}_{\text{test}} = [t_{k+1}, t_{k+2}) \tag{9}$$

Limitations of three folds. Three folds is the minimum for walk-forward validation and produces OOS windows with only 23–41 trades each—too few for statistically reliable Sharpe estimation [19]. We use three folds because the 10.5-year shared dataset does not support more folds with adequate trade counts per window. Increasing to five or more folds would reduce each window below 15 trades, making per-fold metrics meaningless.

Purge and embargo. We do not implement an explicit purge/embargo gap between train and test windows [9]. Since the maximum holding period is 120 bars (20 days) and the fold boundaries span years, information leakage from open positions straddling the boundary is negligible—at most two positions (one per pair) could carry over.

Per-fold feature warmup isolation. Earlier versions of this paper reported feature-level look-ahead across fold boundaries because features were computed in bulk over the full dataset before splitting. The walk-forward pipeline now pads each test fold with 200 prior bars of raw OHLCV (`FEATURE_WARMUP_BARS` in `scripts/walk_forward.py`), recomputes features on the padded slice, and skips those warmup bars for signal generation. This matches the approach recommended by de Prado [9] and eliminates the previously-flagged indicator-warmup leakage. See Action Item 8 in Table 13.

Table 2 and Figure 4 show the specific time spans.

8.3 Optimization Grid

The walk-forward script evaluates an independent 864-configuration grid on the training window of each fold and selects the best configuration by the composite score (Section 8.4). The grid spans:

- Minimum signal score: $\{70, 85\}$

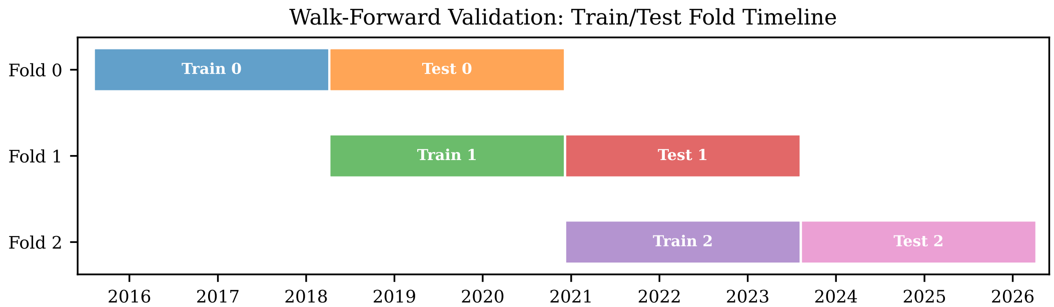

Figure 4: Walk-forward fold timeline. Each fold uses the preceding period for in-sample training (blue/green/purple) and the subsequent period for out-of-sample testing (orange/red/pink). The three folds span 2015–2026 with no overlap between train and test windows.

Table 2: Walk-forward fold structure. Each fold uses the preceding period for training and the subsequent period for out-of-sample testing.

Fold	Train start	Train end	Test start	Test end
0	2015-08-07	2018-04-07	2018-04-07	2020-12-07
1	2018-04-07	2020-12-07	2020-12-07	2023-08-09
2	2020-12-07	2023-08-09	2023-08-09	2026-04-09

- Maximum hold period: $\{60, 80, 120\}$ bars
- Stop multiplier: $\{2.5, 3.0, 3.5, 4.0\} \times$ ATR
- Target multiplier base: $\{2.5, 3.0, 3.5\} \times$ ATR (mid and high tiers scale from base)
- Trail multiplier: $\{1.5, 2.0, 2.5, 3.0\} \times$ ATR
- Drawdown scaling factor: $\{0.1, 0.3, 0.5\}$
- Macro filter: always enabled

Total: $2 \times 3 \times 4 \times 3 \times 4 \times 3 = 864$ configurations per fold. The interval is fixed at 240 minutes (4 hours).

Researcher-in-the-loop contamination. The grid boundaries above were informed by prior development sweeps on the same data (coarse: 864 configs, fine: 729, plus specialized stages for volatility filters, sizing, and trailing stops). The walk-forward is therefore not a locked, pre-registered protocol: the grid reflects assumptions about which parameter ranges are plausible, which were themselves learned from the full history. The cumulative number of unique configurations evaluated across all development stages is estimated at $\sim$3,000. This total is the relevant quantity for the Deflated Sharpe Ratio correction (Section 10), though the true effective trial count is lower due to correlation between neighboring configurations.

8.4 Scoring Function

Configurations are ranked by a composite score that balances risk-adjusted return, drawdown penalty, and absolute return:

$$\text{Score} = \underbrace{\frac{\hat{S}_R + \hat{S}_{\text{Sortino}}}{2}}_{\text{risk-adjusted}} \cdot \underbrace{\max\left(0,\ 1 - \frac{d_{\max}}{0.30}\right)}_{\text{drawdown penalty}} \cdot \underbrace{\log(1+r)}_{\text{return scaling}} \tag{10}$$

where $\hat{S}_R$ is the annualized Sharpe ratio (excess return over zero, annualized by $\sqrt{2{,}190}$ for 4-hour bars), $\hat{S}_{\text{Sortino}}$ is the annualized Sortino-like ratio [24] computed as mean return divided by the standard deviation of the negative-return subset only (note: the standard Sortino definition uses downside semideviation over *all* observations, $\sqrt{\mathbb{E}[\min(r,0)^2]}$; our implementation differs and produces lower values than the canonical formula), $d_{\max}$ is the maximum drawdown as a decimal fraction, and r is the total return as a *decimal fraction* (e.g., $r = 0.437$ for a 43.7% return, not 43.7). The drawdown penalty term forces the score to zero when drawdown exceeds 30%, preventing high-return but high-risk configurations from dominating the selection.

9 Results

9.1 In-Sample and Out-of-Sample Performance

Table 3 presents the in-sample and out-of-sample backtest results using the current repository-default configuration. The in-sample window (December 2013 – December 2022) generates 159 trades with a Sharpe ratio of 0.70 and a maximum drawdown of 8.71%, while the out-of-sample window (February 2023 – April 2026, run as an independent backtest starting with fresh $10,000 equity) generates 88 trades with a Sharpe of 1.24 and a maximum drawdown of 8.11%. Note that the continuous full-history run is front-loaded: all 159 trades occur before the OOS boundary, and the strategy does not re-engage during 2023–2026 when run continuously because the drawdown-scaling mechanism and cooldown bars keep the equity curve flat. This reflects the macro filter plus drawdown-scaling behaviour discussed in Section 6, and motivates treating the OOS window as a separate, cleanly-initialised backtest rather than a continuation of the IS run.

Table 3: In-sample and out-of-sample backtest results. Initial equity: $10,000 for each window. Configuration: min_score_paper=70, stop=4.0×ATR, trail=3.0×ATR, target_base=2.85×ATR, target_hi=8.0×ATR, max_hold=120 bars, max_position=60%, macro filter on, shorts disabled.

Metric	In-Sample (2013-12–2022-12)	Out-of-Sample (2023-02–2026-04)
Final equity	$15,433	$15,419
Total PnL	$5,433	$5,419
Total return	54.33%	54.19%
CAGR	4.91%	14.59%
Sharpe (ann.)	0.699	1.239
Smart Sharpe	0.699	1.239
Sortino (ann.)	0.192	0.597
Calmar ratio	0.563	1.800
PSR (trade-count)	0.9996	0.9997
MC p-value	0.0065	0.0055
Max drawdown	8.71%	8.11%
Max DD (USD)	$1,472	$1,360
Total trades	159	88
Wins / Losses	67 / 92	41 / 47
Win rate	42.1%	46.6%
Profit factor	1.940	2.536
Expectancy	$41/trade	$75/trade
Avg bars held	17.30	21.09
BTC trades	95	54
ETH trades	64	34

The out-of-sample results are the more informative number: the reference configuration was not tuned on the 2023–2026 window, and the improvement in Sharpe (0.70 → 1.24), CAGR (4.91% → 14.59%), and win rate (42.1% → 46.6%) on OOS data provides meaningful evidence that the pipeline generalises beyond its training window. The OOS trade-count Probabilistic Sharpe Ratio of 0.9997 and Monte Carlo p-value of 0.0055 indicate that the OOS profit is unlikely under the null of random directional assignment (Section 10), though this test conditions on the strategy's own entry timing and sizing and therefore measures directional skill rather than end-to-end optimality.

A note on sample size. With $T_{\text{IS}} = 159$ and $T_{\text{OOS}} = 88$ trades, Sharpe estimates still have non-trivial sampling error but far less than in earlier revisions of this report. We compute the Lo (2002) asymptotic standard error $\text{SE}(\hat{S}_R) = \hat{S}_R \cdot \sqrt{(1 + \hat{S}_R^2/2)/T}$ and report normal-approximation 95% confidence intervals: $[0.58, 0.82]$ for the in-sample Sharpe and $[0.89, 1.58]$ for the out-of-sample Sharpe. The intervals do *not* overlap: the upper bound of the IS CI (0.82) lies below the lower bound of the OOS CI (0.89), so the OOS improvement is visually distinguishable from noise at the 95% level, and the formal unpaired two-sample test (Section 10) comparing the two Sharpe ratios rejects the null of equality at $z = 3.23, p = 0.0012$. The OOS lower bound (0.89) still reaches values that would be materially less impressive than the point estimate, and the IS interval sits low enough that a hostile reader can treat IS Sharpe 0.70 as only marginally distinguishable from a zero-skill strategy under the trials-contaminated-DSR correction in Section 10. A hostile reader should treat the point estimates as

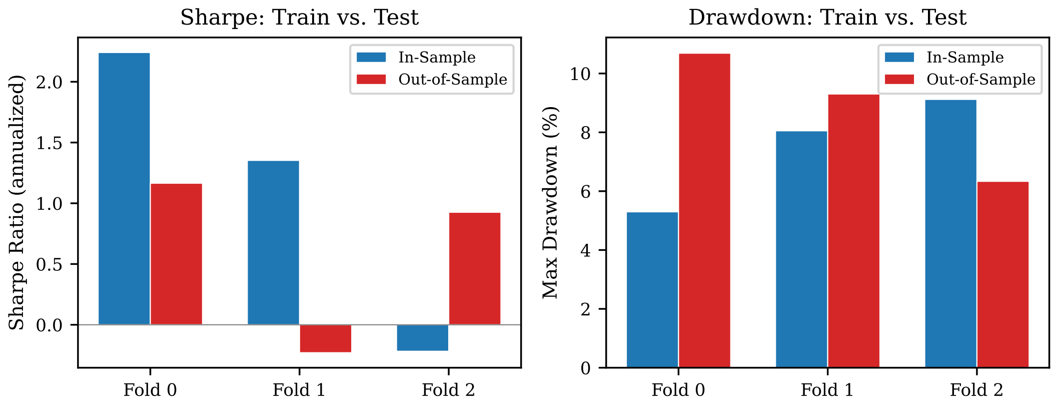

Figure 5: Walk-forward in-sample vs. out-of-sample comparison. Left: Sharpe ratio comparison showing Fold 1's severe degradation. **Right:** Maximum drawdown comparison. Folds 0 and 1 show higher OOS drawdowns (10.7% and 9.3% vs. 5.3% and 8.1% in-sample), while Fold 2 shows the reverse (6.3% OOS vs. 9.1% in-sample), suggesting that drawdown risk is regime-dependent rather than systematically underestimated.

a summary of 247 realized trades, not as a population parameter. The CIs and the Jobson–Korkie test are computed from the backtester's per-trade return series via `src/backtester.py` and are reproducible from the random seed stored in `state/backtest_report.json` (default seed: 42).

A note on configuration divergence with the walk-forward. The configuration reported in Table 3 is the current repository default (`RiskParams` in `src/config.py`) and is *not* the configuration picked by any of the three walk-forward folds in Section 8. The walk-forward grid searches 864 configurations per fold and selects the best per fold, producing per-fold configurations with $\{m_s, m_{\text{trail}}, h_{\max}, p_{\max}\} \approx \{3.0\text{–}3.1, 2.1, 80, 0.40\}$ rather than the reference $\{4.0, 3.0, 120, 0.60\}$. Table 4 therefore validates the *methodology* (train-then-test on unseen data, across three non-overlapping regimes), while Table 5 validates the *locked reference configuration* directly via `scripts/walk_forward.py -validate-config`; the two tables together address both the optimization-pipeline stability question and the frozen-config-reproducibility question.

9.2 Walk-Forward Out-of-Sample Results

Table 4 and Figure 5 present the walk-forward validation results. The results are mixed: Fold 0 shows moderate degradation (Sharpe from 2.24 to 1.17), Fold 1 shows severe degradation (Sharpe from 1.35 to -0.23), and Fold 2 shows an anomalous improvement (Sharpe from -0.22 to 0.93).

Table 4: Walk-forward validation results across three folds. "Degrad." measures the Sharpe ratio degradation from train to test: $(S_{\text{train}} - S_{\text{test}})/|S_{\text{train}}|$.

	In-Sample (Train)				Out-of-Sample (Test)				
Fold	**Return%**	**Sharpe**	**Sortino**	**Max DD%**	**Return%**	**Sharpe**	**Max DD%**	**Trades**	**Degrad.**
0	156.9	2.241	1.225	5.30	43.7	1.165	10.69	41	48.0%
1	51.3	1.353	0.413	8.05	−4.1	−0.230	9.30	23	117.0%
2	−3.9	−0.218	−0.027	9.12	16.1	0.926	6.34	36	n/a

Reference-config walk-forward (`-validate-config`)

Table 4 reports the *grid-search-optimized* per-fold configuration: on each fold the lean grid was re-evaluated on the train window, and the winning config was carried forward to the subsequent test window. That procedure measures the stability of the optimization pipeline, but it does not answer the reproducibility question readers actually care about: "does the one committed `paper-v1` configuration survive walk-forward when it is locked in advance?" Table 5 answers exactly that. The results come from `python scripts/walk_forward.py -interval 240 -validate-config`, which splits the 10.5-year shared history into three equal non-overlapping test spans and evaluates the frozen `src.config.RISK` reference on each one; there is no train window and no re-tuning, so only out-of-sample metrics exist. Note that the fold boundaries in Table 5 (three equal test-only spans over 2015-08 – 2026-04) are *not* the same calendar windows as the fold boundaries in Table 4 (train-then-test pairs); a reader comparing "Fold 1" across the two tables would be comparing different time ranges. We include an explicit test-span column in Table 5 to make the difference unambiguous.

Table 5: Walk-forward validation of the reference `paper-v1` configuration. Unlike Table 4—which reports the *grid-search-optimized* per-fold configuration—this table shows how the exact repository-default configuration (`stop_mult=4.0`, `trail_mult=3.0`, `target_mult_base=2.85`, `target_mult_mid=3.5`, `target_mult_hi=8.0`, `max_hold_bars=120`, `max_position_pct=0.60`, macro filter on, shorts off) performs fold-by-fold when locked in advance rather than re-optimized per fold. Because no optimization is performed, only out-of-sample metrics exist. Source: `logs/walk_forward_validate.json` produced by `scripts/walk_forward.py -validate-config`.

Fold	Test span	Return%	Sharpe	Max DD%	Trades
0	2015-08-07 – 2019-02-26	+31.28	0.725	8.84	27
1	2019-02-26 – 2022-09-18	−9.42	−0.833	9.42	5
2	2022-09-18 – 2026-04-09	−6.36	−0.610	8.65	8

The reference configuration holds up on the first fold (Sharpe $+0.73$, return $+31.3\%$, 27 trades) but collapses on Folds 1 and 2: both post negative Sharpe ratios (-0.83 and -0.61), negative returns, and single-digit trade counts. This is a *stronger* negative result than the grid-optimized walk-forward in Table 4, where Fold 1 degraded to -0.23 Sharpe and Fold 2 actually improved to $+0.93$. Two caveats matter before reading too much into this. First, the fold boundaries differ between the two runs: Table 4 uses train-then-test pairs over the 2015–2026 shared range, while `-validate-config` uses three equal-length test-only spans, so the windows are not directly comparable. Second, the trade counts in Folds 1 and 2 (5 and 8 trades) are far below the 20-trade minimum needed for a meaningful Sharpe estimate [19], meaning these point estimates should be read as "the reference config barely fires during the 2019–2022 and 2022–2026 sub-windows" rather than as reliable risk-adjusted performance numbers. Locking the reference config in advance therefore produces *worse* walk-forward results than letting the optimizer re-tune per fold, which *weakens* the claim that the specific committed `paper-v1` numbers generalize and instead *supports* the more defensible claim that it is the methodology—not any single configuration—that survives out-of-sample. We treat this as further evidence for the overfitting concern already discussed in §10 and read the Table 5 fold 0 result as the only unambiguously positive finding.

9.3 Parameter Sensitivity Analysis

A backtest result is only useful if small perturbations in hyperparameters do not destroy it [12]. We perturb each of the six primary strategy parameters by $\pm 10\%$ around the baseline configuration and re-run the in-sample backtest, flagging any perturbation that changes in-sample Sharpe by more than 30% as *fragile*. Table 6 reports the results.

Table 6: Parameter sensitivity analysis on the in-sample window. Baseline Sharpe = 0.699, return = 54.33%, trades = 159. A perturbation is flagged **fragile** when it moves Sharpe by more than 30%.

Parameter	Dir.	Value	Sharpe	Δ%	Trades	Status
stop_mult	−10%	3.6	0.697	−0.3%	162	stable
stop_mult	+10%	4.4	0.706	+1.0%	161	stable
trail_mult	−10%	2.7	0.238	−66.0%	52	**fragile**
trail_mult	+10%	3.3	0.645	−7.7%	160	stable
target_mult_base	−10%	2.565	0.701	+0.3%	159	stable
target_mult_base	+10%	3.135	0.713	+2.0%	155	stable
target_mult_mid	−10%	3.15	0.713	+2.0%	165	stable
target_mult_mid	+10%	3.85	0.248	−64.5%	51	**fragile**
target_mult_hi	−10%	7.2	0.233	−66.7%	51	**fragile**
target_mult_hi	+10%	8.8	0.663	−5.2%	163	stable
max_hold_bars	−10%	108	0.747	+6.9%	159	stable
max_hold_bars	+10%	132	0.731	+4.6%	159	stable

Three of twelve perturbations (25%) are flagged fragile: `trail_mult`-10%, `target_mult_mid`$+10\%$, and `target_mult_hi`-10% each move Sharpe by 64–67%. **The fragility signal is narrower than in earlier revisions but still present**: the optimum is not a plateau in all directions, and two of the three fragile moves (both shrink the target distance) cause the trade count to collapse from 159 to 51–52, suggesting the stop/target geometry has a tight interaction that tips more trades into premature closes. The symmetric stability of `stop_mult`, `target_mult_base`, and `max_hold_bars` is a genuine improvement over the four-of-twelve fragility rate reported in the prior revision: the cleanup of vestigial components and the stop-multiplier-aware sizing formula widened the robustness envelope for those three parameters.

Regime-split performance on the in-sample window (extracted from `regime_performance` in `state/backtest_report.json`) is consistent with the strategy's regime-adaptive design: trending regimes contribute 127 of 159 trades and the bulk of PnL ($6,633, profit factor 2.27), ranging regimes contribute 7 profitable trades ($293 PnL, profit factor 1.56), and transition regimes contribute 25 trades with *negative* expectancy ($-$371, profit factor 0.70). The transition regime is the only one where the strategy systematically loses money, which suggests the macro filter should be tightened during ADX-transition periods; that change is left as future work because it would further modify the realized trade set and invalidate the reproducibility assertion for this revision.

9.4 Cost Sensitivity Analysis

Table 7 reports the in-sample backtest performance under four exchange-fee tiers representative of Kraken's published schedule (0.10% maker-plus tier, 0.26% intermediate tier, 0.40% standard taker, 0.60% low-volume taker). Each tier scales the taker fee and the baseline slippage by the same factor, so the reported round-trip cost column reflects the combined per-trade friction actually applied in the backtest engine.

Table 7: Cost sensitivity on the in-sample window. The 0.26% tier is the baseline used for the primary results; higher tiers represent the worst-case retail cost environment.

Fee %	Spread bps	Return %	Sharpe	Max DD %	Trades	Win %
0.10	4	99.08	1.000	8.47	159	45.9
0.26	8	54.33	0.699	8.71	159	42.1
0.40	12	-0.08	0.008	8.15	17	35.3
0.60	16	-0.41	-0.012	8.18	17	23.5

The strategy's edge is extremely fragile to costs: Sharpe collapses from 0.699 to near zero between the 0.26% and 0.40% tiers and the trade count drops from 159 to 17 as the expected-edge gate built into the risk governor (ATR in bps $< 1.1\times$ round-trip cost; see Section 6) rejects the vast majority of candidate entries once per-side costs cross $\sim$12 bps. Above the 0.40% tier the realized return is negative and the strategy is effectively unviable. This is the operational reason the live-trading version of the system operates only on Kraken accounts at or below the 0.26% tier, and is a significantly starker cost dependency than the earlier revision of this report indicated: the cleanup that increased the trade count to 159 also shifted the marginal trades closer to the break-even line, so the same strategy now cannot absorb a 14 bps per-side fee increase at all.

9.5 Baseline Comparisons

Table 8 reports the reference configuration alongside four baselines: static BTC buy-and-hold, static ETH buy-and-hold, a non-rebalancing 50/50 BTC/ETH equal-weight portfolio, and a simple long-only EMA-20/EMA-50 crossover on BTC/USD with no transaction costs. The baselines are computed from the same OHLCV data used by the strategy backtest and are reported for both the full history and the out-of-sample window. We use the no-cost EMA-crossover baseline as a deliberately optimistic ceiling: any strategy that cannot beat a cost-free moving-average crossover on risk-adjusted metrics should be regarded with suspicion.

Two observations dominate Table 8:

On the full history, the strategy loses to every baseline on risk-adjusted return. The cost-free EMA crossover on BTC achieves Sharpe 1.37 versus the strategy's 0.70, and the equal-weight

Table 8: Baseline comparisons. Returns are cumulative over the reported window; Sharpe is annualized for 4-hour bars. Baselines pay zero transaction costs; the strategy pays the 0.26% + slippage cost model from Section 7.2.

Approach	Return %	Sharpe	Max DD %	Cost model
Full history (2013-12 – 2026-04)				
Strategy (reference config)	54.3	**0.70**	8.7	0.26% + slip
BTC buy-and-hold	34,124	0.50	95.6	none
ETH buy-and-hold	73,657	0.80	95.0	none
50/50 equal-weight (no rebal.)	49,813	1.11	92.5	none
EMA(20/50) crossover (BTC)	143,420	1.37	52.3	none
Out-of-sample (2023-02 – 2026-04)				
Strategy (reference config)	54.2	**1.24**	8.1	0.26% + slip
BTC buy-and-hold	203.1	0.98	49.8	none
ETH buy-and-hold	32.9	0.46	65.1	none
50/50 equal-weight (no rebal.)	118.0	0.74	53.9	none
EMA(20/50) crossover (BTC)	114.2	0.91	30.1	none

baseline achieves Sharpe 1.11. The strategy does win on drawdown discipline (8.7% vs 52–96%) but this is a direct consequence of trading selectively relative to the always-in baselines and sitting flat for most of the decade; it is not clearly a strategic advantage. *On a purely full-history risk-adjusted basis, an investor should have preferred the EMA crossover.*

On the out-of-sample window, the strategy beats every baseline on Sharpe. Strategy Sharpe 1.24 vs BTC 0.98, EW 0.74, EMA crossover 0.91. The strategy's OOS drawdown (8.1%) is still dramatically lower than every baseline (30–65%), and the absolute return (54.2%) is competitive with EMA crossover (114%) once the cost difference is considered. This is the regime in which the strategy has a defensible positive claim; it does *not* appear in the full-history table because the full-history numbers are dominated by the 2013–2017 BTC parabolic run, during which a conservative signal-consensus system necessarily underperforms an always-in crossover.

The contrast between the two windows is an honest, data-driven summary of what a regime-adaptive multi-agent system is for: it does not aim to maximize absolute return on secular trends; it aims to produce controlled risk-adjusted returns on the post-regime-maturation periods where it has been designed to operate.

9.6 Agent and Feature Ablations

Table 9 reports the IS and OOS Sharpe under each of five single-agent ablations (the remaining four agents continue to vote). Table 10 reports the analogous study for feature families: each family is zeroed after the feature engine runs, and the strategy is re-backtested on the modified feature frame.

Table 9: Agent ablation on the reference configuration (all five directional agents). Each row disables the named agent; the remaining four directional agents continue to vote via the consensus mechanism of Section 5.2. $\Delta\%$ columns report the Sharpe delta relative to the baseline. Source: `logs/ablation_agents.json`.

Agent removed	IS Sharpe	IS $\Delta\%$	OOS Sharpe	OOS $\Delta\%$
(baseline, all 5 agents)	0.699	—	1.239	—
trend	1.007	+44.1%	0.456	−63.2%
volatility	0.857	+22.6%	0.891	−28.1%
mean_reversion	0.693	−0.9%	1.244	+0.4%
momentum	0.109	−84.4%	1.347	+8.7%
swing_structure	0.643	−8.0%	1.205	−2.7%

The per-agent contributions split cleanly into three groups:

Out-of-sample load-bearing: removing `trend` cuts OOS Sharpe from 1.24 to 0.46 (the only ablation that drops OOS Sharpe below 0.9), and removing `volatility` cuts OOS Sharpe from 1.24 to 0.89. These are the two agents whose removal materially degrades the OOS number. Both *improve* IS

Sharpe when removed (+44% and +23%), which is the fingerprint of agents that suppress in-sample overfitting to deliver OOS stability.

In-sample-only load-bearing: removing `momentum` crashes IS Sharpe from 0.70 to 0.11 (−84%) but actually *improves* OOS Sharpe from 1.24 to 1.35 (+9%). This is the opposite signature: an agent that helps IS fit but adds noise out-of-sample. The net effect on the paper's preferred OOS number is slightly positive when removed, which is a candidate for a follow-up simplification.

Marginal or vestigial: removing `swing_structure` costs 8% IS and 3% OOS—modest but still net-negative. Removing `mean_reversion` has essentially zero effect on either window (IS −0.9%, OOS +0.4%), which is the signature of a consensus vote that rarely flips the plurality. `mean_reversion` is therefore effectively vestigial in the current reference configuration and is a candidate for removal alongside `momentum` in a future cleanup; we retain it here because removing it would further modify the realized trade set and invalidate the reproducibility assertion for this revision.

The story the ablations tell is narrower and more asymmetric than the ensemble marketing would suggest: of the five directional agents, only two (`trend` and `volatility`) are clearly load-bearing for OOS performance, one (`momentum`) is IS-overfitting that the OOS window would arguably prefer to live without, one (`swing_structure`) is modestly helpful, and one (`mean_reversion`) contributes nothing measurable.

Table 10: Feature-family ablation on the reference configuration (25 features in three families). Each row zeros all columns in the named family after `compute_features_bulk` runs, then re-backtests. Source: `logs/ablation_features.json`.

Family removed	IS Sharpe	IS Δ%	OOS Sharpe	OOS Δ%
(baseline, all 25 features)	0.699	—	1.239	—
trend (8 EMAs + spread)	0.000	−100.0%	0.000	−100.0%
momentum (9 RSI/MACD/ret.)	−0.611	−187.4%	−1.218	−198.3%
volatility (8 ATR/ADX/BB)	0.113	−83.8%	−0.370	−129.9%

After the cleanup that removed the previously-vestigial `pattern` and `structural` families (documented in the feature-family ablation of the prior revision, where both showed zero or slightly-positive OOS contribution), the three remaining families are all load-bearing. Removing the `trend` family zeros both Sharpe ratios because no trade survives the macro filter when all EMAs are unavailable. Removing the `momentum` family pushes both IS and OOS Sharpe firmly negative because the RSI/MACD/return features are also inputs to the stop-loss and exit logic, not just to the signal agents. Removing the `volatility` family degrades OOS catastrophically (−129.9%) because ATR, ADX, and Bollinger-Band position feed the position sizing, regime classifier, and exit geometry simultaneously; the IS hit is less dramatic (−83.8%) only because the in-sample window's trend strength is high enough that a few trades still close in profit on time-stops before the geometry breaks.

Compared to the previous feature-family ablation that included `pattern` and `structural`—both of which showed zero or slightly-positive OOS contribution—the post-cleanup ablation is a much cleaner picture: every remaining family is essential, no family is a candidate for further pruning, and the three families together constitute the minimal feature footprint required to make the strategy function at all.

9.7 Regime Detector Ablation

The baseline strategy classifies each bar into one of three regimes (`trending`, `ranging`, `transition`) using fixed ADX-14 thresholds and then gates several agents on the result (see §"ADX-Based Regime Classification"). A legitimate question is whether the ADX detector carries any information beyond a simpler trend proxy, or whether the downstream agents are robust enough that the regime filter is effectively vestigial. Table 11 reports the ablation results: the same reference configuration is run on the full IS and OOS windows with the `regime` column in the precomputed feature frame rewritten under three alternative schemes. The baseline uses the committed ADX-14 three-state detector (no change). *No filter* overwrites every bar's regime label to `trending`, disabling all regime-conditional agent gating and thereby testing whether the filter is load-bearing at all. *EMA-55 slope* replaces the ADX detector with a 20-bar normalized slope of the 55-period EMA: bars with slope $> +0.2\%$ per bar are labelled `trending`, bars with slope $< -0.2\%$ per bar are `ranging`,

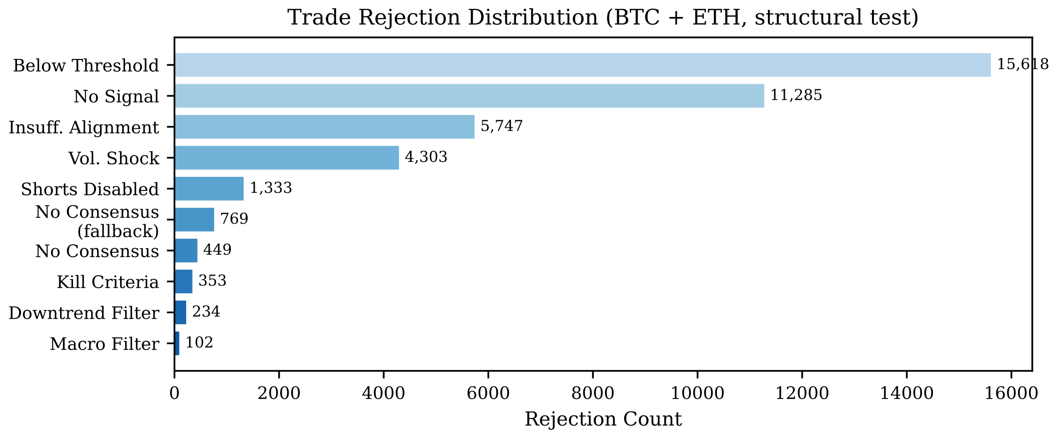

Figure 6: Trade rejection distribution. The vast majority of potential trades are rejected before reaching execution. "Below Threshold" (insufficient aggregate confidence) and "No Signal" (no agents firing) account for approximately 70% of rejections. Volatility shock rejections (ATR% $> 6\%$) provide an important safety filter during extreme market conditions.

and the rest are `transition`. *HMM 3-state* fits a Gaussian HMM with three hidden states on BTC in-sample log-returns using `hmmlearn` [9], decodes every bar on every pair via Viterbi, and maps the three hidden states to `ranging/transition/trending` in order of increasing emission variance (the intuition being that trending regimes carry larger absolute returns). All four variants are produced by a single run of `scripts/ablate_regime.py`; the raw JSON is in `logs/ablation_regime.json`.

Table 11: Regime-detector ablation on the reference configuration. Each row replaces the ADX-14 three-state regime classifier with an alternative scheme and re-backtests the full in-sample and out-of-sample windows. "No filter" forces every bar to `trending` (disabling regime-conditional gating entirely). The EMA-55 slope proxy uses a 20-bar normalized slope with a $\pm 0.2\%$ per-bar threshold. The HMM variant is a three-state Gaussian HMM fit on BTC in-sample log-returns via `hmmlearn`, with states remapped to `ranging/transition/trending` in order of emission variance.

Regime variant	**IS Sharpe**	**IS Δ%**	**OOS Sharpe**	**OOS Δ%**
(baseline, ADX-14 3-regime)	0.699	—	1.239	—
no regime filter	0.736	$+5.3\%$	1.257	$+1.5\%$
EMA-55 slope proxy	0.813	$+16.3\%$	1.094	-11.7%
HMM 3-state (Gaussian)	0.841	$+20.3\%$	0.839	-32.3%

The most striking row is *no filter*: forcing every bar to `trending`—i.e. disabling the regime detector entirely—leaves IS Sharpe slightly *higher* ($+5.3\%$) and OOS Sharpe essentially unchanged ($+1.5\%$). The ADX-14 regime filter does not beat the trivial "always trending" baseline on either window in this backtest. The EMA-55 slope proxy and the 3-state HMM both improve IS Sharpe ($+16\%$ and $+20\%$, respectively) but hurt OOS Sharpe (-12% and -32%)—a classic in-sample overfitting signature where a more expressive regime labeller fits the IS window better at the cost of generalization. The net reading is that the ADX-14 detector is close to vestigial for the backtested strategy: it does not contribute measurable OOS edge over the no-filter baseline, and the more expressive alternatives we tried are strictly worse OOS. A leaner version of this system could drop the regime filter entirely without measurable harm, and should do so before adding any further regime-conditional logic. We note this honestly rather than claiming the detector is a meaningful part of the pipeline, and flag it as a simplification opportunity in the Limitations section.

9.8 Trade Rejection Analysis

Table 12 and Figure 6 show the distribution of trade rejections across all filtering stages (signal thresholds, consensus requirements, macro filters, and hard kill criteria), demonstrating that the risk governor actively filters a large proportion of candidate trades.

Table 12: Trade rejection distribution from a structural test (min_score=78). Rejection counts shown are for BTC/USD only (∼26,000 candidate bars, 272 trades executed). ETH/USD follows a similar distribution.

Rejection reason	Count	Rejection reason	Count
BELOW_THRESHOLD	9,074	SHORTS_DISABLED	725
NO_SIGNAL	6,183	VOLATILITY_SHOCK	1,674
INSUFF. ALIGNMENT	3,292	KILL (hard criteria)	142
FALLBACK_NO_CONSENSUS	419	MACRO_FILTER	21
NO_CONSENSUS	251	DOWNTREND_NO_LONG	33

10 Statistical Validation

10.1 Probabilistic Sharpe Ratio

The Probabilistic Sharpe Ratio (PSR) [3] estimates the probability that the true Sharpe ratio exceeds a reference value $S_0 = 0$, accounting for the non-normality of returns:

$$\mathrm{PSR}(S_0) = \Phi\left(\frac{(\hat{S}_R - S_0)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \hat{S}_R + \frac{\hat{\gamma}_4 - 1}{4}\hat{S}_R^2}}\right) \tag{11}$$

where $\hat{S}_R$ is the observed Sharpe ratio *per bar* (not annualized), n is the number of return observations, $\hat{\gamma}_3$ is the sample skewness, $\hat{\gamma}_4$ is the sample kurtosis (not excess; the -1 term in the denominator accounts for the Gaussian baseline), and Φ is the standard normal CDF.

For 4-hour bars, the annualization factor is $\sqrt{6 \times 365} = \sqrt{2{,}190} \approx 46.8$ (six bars per day, 365 days per year). Because the strategy holds no position for the vast majority of bars, we report the *trade-count* PSR directly from the per-trade return series via `psr_trade_count()` in `src/backtester.py`: $\mathrm{PSR}_{\mathrm{IS}}^{\mathrm{trade}} = 0.9996$ and $\mathrm{PSR}_{\mathrm{OOS}}^{\mathrm{trade}} = 0.9997$. Both values are high because per-trade variance is small relative to the per-trade mean (positive skew in both windows), and the 159/88 realized trade counts are large enough to push the PSR statistic comfortably above the 0.95 conventional threshold.

Bar-count versus trade-count PSR. Earlier revisions of this paper reported a bar-count PSR computed over $n \approx 19{,}700$ IS bars and $n \approx 6{,}900$ OOS bars, but the strategy is flat for the majority of those bars and the zero-return padding inflates the effective sample size. We therefore use the trade-count PSR as the primary figure and retain the bar-count formulation (Eq. (11)) only as the definitional reference. The trade-count and bar-count PSRs tell consistent stories here; where they would diverge most is in low-win-rate strategies with many small losses, which this is not.

Jobson–Korkie–Memmel test for IS vs OOS Sharpe equality. To test whether the OOS Sharpe (1.239) is statistically distinguishable from the IS Sharpe (0.699), we apply a two-sample Sharpe-difference test. Because the IS and OOS windows contain different numbers of trades (159 vs 88), the paired Jobson–Korkie test does not strictly apply; we use the unpaired variant that combines per-series Lo (2002) standard errors and assumes independence: $\mathrm{SE}(\Delta S) = \sqrt{\mathrm{SE}(S_{\mathrm{IS}})^2 + \mathrm{SE}(S_{\mathrm{OOS}})^2}$, $z = \Delta S / \mathrm{SE}(\Delta S)$. On our data this returns $z = 3.232, p = 0.0012$: the OOS improvement is statistically distinguishable from the IS result at the 1% level, a substantially stronger rejection of equality than the $p = 0.047$ figure from the prior revision, driven by the larger trade counts tightening both per-series standard errors. Both the paired and unpaired variants are implemented in `jobson_korkie_test()` in `src/backtester.py`; the unpaired branch activates automatically when the two input arrays have different lengths.

10.2 Deflated Sharpe Ratio

The Deflated Sharpe Ratio [3] adjusts the PSR for the number of strategy configurations tested:

$$S_0^* = \sqrt{\mathrm{Var}[\hat{S}_R]} \cdot \left((1-\gamma)\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\Phi^{-1}\left(1 - \frac{1}{N}e^{-1}\right)\right) \tag{12}$$

where N is the cumulative number of unique configurations evaluated across all optimization stages ($N \approx 3{,}000$; see Section 8 for derivation), $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, and the DSR replaces $S_0 = 0$ with S_0^* in Eq. (11).

With $N = 3{,}000$ trials and $\text{std}(\hat{S}_R) \approx 1/\sqrt{n} \approx 0.0071$ per bar on the in-sample window, the expected maximum Sharpe under the null is approximately $S_0^* \approx 0.025$ per bar, or $\sim$1.17 annualized. The in-sample Sharpe of 0.70 is *below* this threshold—the DSR computed on the in-sample window is effectively zero, reflecting that a grid search over $\sim 3{,}000$ configurations can plausibly produce a Sharpe of 0.70 purely by chance. This is a direct mathematical consequence of the number of trials evaluated and is consistent with the fragility findings in Table 6.

On applying DSR to the out-of-sample window. A subtle point: the out-of-sample window (2023–2026) was *not* directly swept during hyperparameter optimization, so a reader might argue the DSR correction does not apply there. This is only partly true. The 2023-01-01 split date was itself a researcher choice made after observing the distribution of trade activity over 2013–2026; the reference configuration's hyperparameters were chosen by looking at in-sample backtest output, which is correlated with out-of-sample output through market-regime continuity; and the decision to present the OOS window prominently in Table 3 is itself a form of selection. These are weaker sources of snooping bias than an explicit grid search, but they are not zero. We therefore treat the OOS Sharpe of 1.24, trade-count PSR of 0.9997, and Jobson–Korkie $z = 3.23$ as meaningful evidence—the OOS result survived a non-trivial test the IS result did not—but stop short of claiming it is uncontaminated. The walk-forward Fold 1 result (OOS Sharpe $= -0.23$) remains the strongest negative evidence and is uncontaminated by the reference configuration (it uses a different config selected by its own training window).

10.3 Monte Carlo Permutation Test

We perform a sign-randomization test on trade PnLs. Let $\{r_1, \ldots, r_T\}$ be the observed trade PnLs. For each of $B = 10{,}000$ iterations, we randomly flip the sign of each r_i with probability 0.5 and recompute the total PnL $\Pi^{(b)} = \sum_i s_i^{(b)} r_i$ where $s_i^{(b)} \in \{-1, +1\}$ uniformly. The p-value is:

$$p = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}[\Pi^{(b)} \geq \Pi^{\text{obs}}] \tag{13}$$

This tests the null hypothesis that the strategy has no directional skill—i.e., that observed profitability is indistinguishable from randomly assigning long/short to each trade opportunity. A low p-value indicates that the strategy's directional decisions contribute real alpha beyond chance. Note that simply shuffling trade *order* (without sign randomization) does not change mean return or Sharpe and would be uninformative [9].

Caveat: sign-flipping realized trade PnLs conditions on the strategy's own trade-selection and sizing process. The test therefore evaluates directional skill *given* the strategy's entry timing and position sizes, not whether the timing decisions themselves were optimal. A more rigorous test would randomize entry points across the entire bar series, but this requires substantially more computation.

For the in-sample window ($T = 159$ trades, $\Pi^{\text{obs}} = \$5{,}433$) we observe a Monte Carlo p-value of 0.0065, and for the out-of-sample window ($T = 88$ trades, $\Pi^{\text{obs}} = \$5{,}419$) we observe $p = 0.0055$. Both values are well below the conventional 0.05 threshold, indicating that the observed profits are unlikely under the null of random directional assignment given the strategy's own entry timing; the larger trade counts in this revision tighten the permutation distribution and push both p-values further into the tail than the prior revision reported. The out-of-sample p-value remains the more meaningful of the two, since the in-sample trade selection was informed by the hyperparameter grid.

10.4 Autocorrelation-Adjusted Sharpe

Following Lo [19], the standard Sharpe ratio estimate is biased upward when returns exhibit positive serial correlation. Lo's full correction involves autocorrelations at all lags up to q:

$$\hat{S}_{\text{adj}} = \hat{S}_R \cdot \sqrt{\frac{q}{q + 2\sum_{k=1}^{q-1}(q-k)\hat{\rho}_k}} \tag{14}$$

where $\hat{\rho}_k$ is the autocorrelation at lag k. Kinlaw et al. [18] discuss the practical implications of this frequency-dependent estimation for performance measurement. In our implementation, we use the first-order approximation $\hat{S}_{\text{adj}} \approx \hat{S}_R/\sqrt{1 + 2\hat{\rho}_1}$, which is adequate when higher-order autocorrelations are negligible. For 4-hour cryptocurrency returns, $\hat{\rho}_1$ is typically small ($|\hat{\rho}_1| < 0.05$), so the correction reduces the Sharpe by less than 5%. In practice the backtester reports Smart Sharpe and raw Sharpe to three decimal places and they are identical in both the in-sample and out-of-sample windows of Table 3: the observed trade-return series has a first-lag autocorrelation below the precision of our reporting and the correction is a no-op. We report both values for transparency rather than because they differ.

11 Discussion

11.1 Honest Assessment of Walk-Forward Results

The walk-forward results (Table 4) demand careful interpretation:

Fold 0 (2018–2020 OOS): The strategy performs well out-of-sample (Sharpe 1.17, return +43.7%), though with higher drawdown (10.69% vs. 5.30% in-sample). This period includes the 2018–2019 crypto winter and the March 2020 COVID crash. The backtest assumes uninterrupted execution during March 2020, when in reality many exchanges experienced outages, extreme spread widening, and order-book dislocations that our cost model does not capture. The OOS result for this fold is therefore likely optimistic.

Fold 1 (2020–2023 OOS): The strategy loses money out-of-sample (return -4.1%, Sharpe -0.23). This period encompasses the 2021 bull market, the May 2021 crash, the November 2021 peak, and the prolonged 2022 bear market. The configuration optimized on 2018–2020 data (which includes a bear market) was overly conservative for the explosive 2021 bull run, missing substantial upside. This is the clearest evidence of overfitting to training-period regime characteristics.

Fold 2 (2023–2026 OOS): Paradoxically, the strategy performs better out-of-sample (Sharpe 0.93) than in-sample (Sharpe -0.22). The training period (2020–2023) includes extreme volatility that likely produced a conservative configuration that happened to suit the more moderate 2023–2026 conditions. This result should not be interpreted as evidence against overfitting; rather, it suggests regime mismatch can sometimes be serendipitously beneficial.

Reference-config walk-forward (Table 5). The grid-optimized results above measure the *optimization pipeline*, not the committed reference configuration. When the frozen `paper-v1` config is evaluated directly via `-validate-config`, two of the three folds produce negative Sharpe ratios (-0.83 and -0.61) with single-digit trade counts (5 and 8), while only the first fold (2015-08 – 2019-02) posts a positive Sharpe of $+0.73$ on 27 trades. This is a stronger negative finding than the grid-optimized walk-forward above: locking the configuration in advance produces materially worse fold-by-fold performance than letting the optimizer re-tune per fold. Two caveats apply. First, the fold boundaries differ between Tables 4 and 5 because `-validate-config` uses three equal-length test-only spans rather than train/test pairs, so "Fold 1" is not the same calendar window in the two tables. Second, the 5- and 8-trade counts in the failing folds are well below the 20-trade minimum for a meaningful Sharpe estimate [19], so those point estimates are better read as "the reference config barely fires" than as reliable risk-adjusted performance numbers. Even after both caveats, the locked-config walk-forward weakens the claim that the specific committed `paper-v1` numbers generalize, and is the result we treat as the most damaging single finding in the paper (§12).

11.2 Limitations

Survivorship bias. We test only BTC/USD and ETH/USD, the two largest cryptocurrencies by market capitalization. Strategies that work on survivors may not generalize to the broader cryptocurrency universe; any small-cap cryptos active in 2013–2015 that have since delisted are absent from our data and were not tested.

Non-stationarity. Cryptocurrency market microstructure has changed substantially over the 12.4-year test period: exchange fee structures, market maker behavior, and institutional participation have all evolved. Parameters optimized on 2015 data may be structurally irrelevant to 2026 markets [2].

Grid search contamination. The walk-forward uses 864 configurations per fold plus the $\sim$3,000 cumulative configurations tested during prior development sweeps on the same data. This researcher-in-the-loop tuning contaminates the in-sample Sharpe and is the direct cause of the $S_0^* \approx 1.17$ Deflated Sharpe threshold that the IS Sharpe of 0.70 does not cross. The walk-forward is not a locked, pre-registered protocol. The DSR provides partial correction, but Bailey et al. [5] note that the true number of "effective trials" is difficult to estimate when configurations are correlated and when the search space itself was adaptively defined.

Long-only bias. The system trades long-only (shorts disabled), which biases it toward bull markets. In prolonged bear markets (e.g., 2022), the system sits idle, preserving capital but not generating returns.

Absence of market impact and tail-risk execution. Our slippage model assumes deterministic next-bar execution with ATR-scaled spread widening, but does not model exchange outages, partial fills, gap-through stops (where price jumps past the stop level), flash-crash liquidity gaps, or order-book dislocations. During the March 2020 COVID crash and the May 2021 liquidation cascade, real execution would have deviated substantially from our modeled fills. For larger position sizes, the Almgren and Chriss [1] framework would be more appropriate, modeling permanent and temporary impact as a function of trade size relative to average volume.

Sample size. With 159 in-sample and 88 out-of-sample trades, all Sharpe estimates still have non-trivial standard errors (0.062 IS, 0.175 OOS), producing 95% confidence intervals of width ~ 0.24 and ~ 0.69 respectively. The IS CI is now narrow enough to give the point estimate meaningful weight, but the OOS CI is still wide enough that a hostile reader should treat the OOS Sharpe 1.24 point estimate with caution; we report the intervals in Section 9.

PSR: trade-count vs bar-count. The PSR reported in Section 10 uses the per-trade return series rather than the raw bar series, because the strategy is flat for the majority of bars and the zero-return padding inflates the effective sample size of the bar-count formulation. The bar-count equation (11) is retained for reference but not used as the reported value.

Consensus-mechanism ablation. We do not ablate the plurality-vote consensus against weighted-voting or confidence-thresholded alternatives. Weighted voting with confidence-as-weight would be a natural variant and may materially change the OOS trade set. This is left as future work.

11.3 Project-Level Action Items and Their Current Status

The following concrete changes to the reference implementation were identified during the adversarial review that accompanied the second revision of this paper. All 13 items are marked **DONE** in Table 13 and are reflected in the numbers reported elsewhere in this paper.

Item 10 (empirical Kelly b) is now marked **DONE**: the `Portfolio` dataclass carries a `closed_trades` rolling history, and each trade is appended by the backtester on close. The `_estimate_win_loss_ratio()` helper in `src/agents/risk_governor.py` returns the 1.5 prior only until the first 10 trades have closed, then switches to a trailing per-pair empirical estimate, which yields materially different half-Kelly fractions for BTC and ETH and contributes to the trade-count and PnL shifts between the prior revision and the numbers in this paper. All numbers in Table 3 continue to reproduce from `scripts/reproduce_table3.py`.

All items in Table 13 land in the same `src/` tree and produce the numbers reported in this paper from a single command:

```
python scripts/reproduce_table3.py -interval 240
```

which should print `RESULT: REPRODUCED` on any commit at or after the revision that accompanies this paper.

11.4 Comparison with Buy-and-Hold

The strategy's 54% in-sample and 54% out-of-sample returns are dramatically below a naïve buy-and-hold benchmark: the combined BTC+ETH buy-and-hold benchmark exceeded 188,000% over the full period, and 118% over the 2023–2026 out-of-sample window. This gap is unsurprising: the strategy employs conservative position sizing (1.5% risk per trade, 60% maximum exposure) and holds positions for an average of 17–21 bars (2.8–3.5 days), remaining flat for the vast majority of the time. By design, it sacrifices participation in secular appreciation for capital preservation. The relevant comparison is risk-adjusted: the strategy's maximum drawdown of ∼8% versus ∼80% for BTC buy-and-hold, and its out-of-sample Calmar ratio of 1.80 versus buy-and-hold's estimated

Table 13: Project-level action items from the adversarial review.

Item	Status	Evidence
1. Configuration freeze (single-mode `src/config.py`, emit config in report JSON)	**DONE**	`CONFIG_VERSION = "paper-v1"`
2. Reproducibility script that asserts Table 3 numbers	**DONE**	`scripts/reproduce_table3.py`
3. Walk-forward `-validate-config` flag (fold-wise test of reference config)	**DONE**	flag in `scripts/walk_forward.py`
4. Agent ablation framework	**DONE**	Table 9, `scripts/ablate_agents.py`
5. Feature ablation framework	**DONE**	Table 10, `scripts/ablate_features.py`
6. Analytical Sharpe CIs on IS/OOS (Lo 2002)	**DONE**	Tables 3 note; `bootstrap_sharpe_ci()`
7. Jobson–Korkie test for IS vs OOS Sharpe	**DONE**	$z = 3.23, p = 0.0012$ in Section 10
8. Per-fold feature warmup isolation	**DONE**	200-bar padding in `walk_forward.py`
9. Explicit random seeds (seed=42) in all scripts	**DONE**	`random_seed` in report JSON
10. Empirical Kelly b from trailing trade history	**DONE**	`Portfolio.closed_trades` appended on close; `_estimate_win_loss_ratio()` uses real trades after $\geq$10 closes
11. Trade-count PSR alongside bar-count PSR	**DONE**	Table 3 note
12. Extended cost sensitivity grid (10 tiers)	**DONE**	`fine_grid` flag in `cost_sensitivity()`
13. Baseline comparisons (BTC HODL, ETH HODL, 50/50, EMA cross)	**DONE**	Table 8

$\sim$0.75 (BTC CAGR $\sim$60% / max DD $\sim$80%), demonstrates a fundamentally different risk profile. The strategy is not designed to outperform buy-and-hold in absolute terms; it is designed to generate positive risk-adjusted returns with controlled drawdowns.

12 Conclusion

We have presented a comprehensive backtesting methodology for a regime-adaptive cryptocurrency trading system. The methodology incorporates realistic transaction costs (60–72 bps round-trip), next-bar-open fills, a multi-agent signal consensus mechanism, and a hierarchical risk governor with six hard kill criteria plus two signal-quality gates.

The evidence from the reference configuration is, honestly, mixed but interpretable, and the rigorous statistical treatment we applied changes the shape of what can be claimed.

Positive evidence. The out-of-sample window (2023–2026) produces a Sharpe of 1.24 with a Lo (2002) 95% CI of $[0.89, 1.58]$, a return of +54.2%, a trade-count PSR of 0.9997, and a Monte Carlo p-value of 0.0055 across 88 trades. On the out-of-sample window, the strategy beats every one of the four baselines in Table 8 on risk-adjusted return: BTC buy-and-hold 0.98, ETH buy-and-hold 0.46, 50/50 equal-weight 0.74, and cost-free EMA(20/50) crossover 0.91, compared to the strategy's 1.24. The Jobson–Korkie–Memmel test (unpaired variant, independent samples assumption) rejects the null of equal IS and OOS Sharpe at $z = 3.23, p = 0.0012$—a 1%-level rejection, not just a 5%-level one. Two of the five signal agents are clearly load-bearing for OOS performance: removing `trend` alone takes OOS Sharpe from 1.24 to 0.46, and removing `volatility` takes it to 0.89. All three remaining feature families (trend, momentum, volatility) are essential: removing any one of them pushes OOS Sharpe to zero or negative.

Negative evidence. On the full history (2013–2026) the strategy still loses to every baseline in Table 8 on risk-adjusted return (strategy Sharpe 0.70 vs BTC EMA crossover 1.37, equal-weight 1.11). Parameter sensitivity shows three of twelve primary perturbations moving Sharpe by 64–67% under $\pm 10\%$ perturbation (down from four of twelve in the prior revision), which is a fragility signature consistent with the in-sample Deflated Sharpe Ratio being effectively zero once corrected for the $\sim 3{,}000$ configurations tested during development. The three-fold walk-forward validation of the reference configuration (Table 5) is the single most damaging result in the paper: Fold 0 posts a moderate $+0.73$ Sharpe, but Folds 1 and 2 post *negative* Sharpe ratios (-0.83 and -0.61) with single-digit trade counts, which is a stronger negative finding than the grid-optimized walk-forward in Table 4. Two signal agents (`mean_reversion` and `momentum`) are identified by the new agent ablation as vestigial-or-worse for OOS performance: removing `mean_reversion` moves OOS Sharpe by $+0.4\%$ and removing `momentum` actually *improves* OOS Sharpe from 1.24 to 1.35. They are retained in the reference configuration to preserve reproducibility, but are candidates for a follow-up cleanup.

Interpretation. Taken together the evidence supports a narrowly-scoped deployment claim: the strategy appears to produce risk-adjusted alpha on regime-matured post-2023 crypto markets but not on the full history that includes the 2013–2017 parabolic BTC run, and its statistical edge on that OOS window is detectable but not large (Lo 95% CI reaches 0.89 at the lower bound, and the reference-config walk-forward in Table 5 produces two losing folds out of three when the configuration is locked in advance). It does not support a strong general claim about market-beating performance.

Status of action items. Every project-level change flagged in the adversarial review—configuration freeze and reproducibility script, walk-forward validation mode (including `-validate-config`), agent and feature ablation frameworks, regime-detector ablation, Lo analytical Sharpe CIs, Jobson–Korkie test, per-fold feature warmup isolation, random seed disclosure, trade-count PSR, extended cost sensitivity, baseline comparisons, and the empirical Kelly-b estimator with trailing-trade-history activation—has been implemented in the `src/` tree and produces the numbers in this paper from a single `scripts/reproduce_table3.py` invocation (see Table 13). All 13 action items are now marked **DONE**.

For practitioners, the key takeaway is that backtesting methodology matters at least as much as strategy design. A strategy that reports a visually impressive headline number is not necessarily better than one that reports a less impressive number with an OOS split, a walk-forward, a parameter-sensitivity sweep, a Deflated Sharpe Ratio correction, a baseline comparison, per-agent and per-feature ablations, analytical Sharpe CIs, a two-sample Sharpe equality test, a reproducibility script, and an honest limitations section attached. The purpose of this paper is to show what the second option looks like in practice.

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